

Eigenvalues and inverse problems

Where a symmetric stochastic matrix can be spectrally unique

Difficulty: challenging **Importance:** interesting to specialist**Status:** Open

Rating rationale: Challenging reflects the geometry of isospectral stochastic families in arbitrary dimension; specialist impact concerns the narrow property of spectral uniqueness within that class.

Problem statement

For $n \geq 4$, put $\mathcal{S}_n = \{A \in \mathbb{R}^{n \times n} : A = A^T, A \geq 0, A\mathbf{1} = \mathbf{1}\}$ and $C_n = (\mathbf{1}\mathbf{1}^T - I)/(n-1)$. A matrix $A \in \mathcal{S}_n$ is *spectrally unique* if every $B \in \mathcal{S}_n$ with the same eigenvalues, including multiplicities, satisfies $B = R^T A R$ for a permutation matrix R . Let $[X, Y] = \{(1-t)X + tY : 0 \leq t \leq 1\}$.

Prove or disprove the following necessary condition: every spectrally unique $A \in \mathcal{S}_n$ with $\text{tr } A > 0$ belongs to

$$[I, C_n] \cup \bigcup_{V \in \text{vert}(\mathcal{S}_n)} ([I, V] \cup [C_n, V]).$$

Here a vertex is an extreme point of the indicated convex polytope; it need not be a permutation matrix. Only the stated implication is asserted. This asks where recovering a nonnegative symmetric stochastic matrix from its spectrum can be unique up to relabeling, a different issue from mere spectral feasibility.

References

Mourad and Abbas, [2013 preprint](#), definitions in §1 and Conjecture 5.1, p. 10; [published article](#), *Linear and Multilinear Algebra* 63 (2015), 869–881.

Earlier status check — 2026-09-08

Searches for the exact title, *symmetric doubly stochastic Conjecture 5.1*, and author/title combinations with *counterexample* and 2026 found no resolution. The source solves order three, which is excluded from the remaining statement above.

Audit update — 2026-09-10

Rechecked [Mourad–Abbas](#), §5, Conjecture 5.1, against the trace and locus restrictions here. Searches for spectrally unique symmetric stochastic matrices and later work on that conjecture found no general answer. The order-three classification is outside the displayed unresolved dimensions, so it does not change this entry's status. Evidence remains historical rather than a recent explicit reaffirmation.